

SLOPPY INFERENCES - Learning Study - 20190924 (#28306)

Author(s)

This pre-registration is currently anonymous to enable blind peer-review.
It has one author.

Pre-registered on: 09/24/2019 08:33 AM (PT)

1) Have any data been collected for this study already?

No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?

In an investment game in which there is no relationship between cues (the projects' characteristics) and outcomes (the projects' payoffs), we hypothesize that the slope of the payoffs over time (increasing vs. staying flat vs. decreasing) will exert a significant impact on people's subjective sense of learning.

In particular, we expect that people will be more (less) confident that they have identified the factor(s) driving the projects' performance when the payoffs are increasing (decreasing) rather than constant over time.

3) Describe the key dependent variable(s) specifying how they will be measured.

Bet Amount: After playing the game, participants see three projects with their attributes, and will indicate which project they believe has the highest payoff. They can then bet money (between 0 and 10 cents) on their answer being correct. The bet amount constitutes the dependent variable.

Sense of Learning: Participants will answer 5 questions measuring different aspects of their subjective sense of learning, measured on a 7-point scale. Those items will be appropriately reverse-coded and averaged to form a synthetic measure of subjective learning, which constitutes the dependent variable.

4) How many and which conditions will participants be assigned to?

Participants will be randomly assigned to one of three cells of a one-factor design. We will manipulate the slope of the payoffs so that they decrease vs. stay flat vs. increase over time. The payoffs will be identical for all participants within any specific condition.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

Bet Amount: OLS regression with two dummies for the conditions: "Flat vs. Decreasing" and "Increasing vs. Decreasing".

Sense of Learning: OLS regression with two dummies for the conditions: "Flat vs. Decreasing" and "Increasing vs. Decreasing".

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

No participant will be excluded from the study.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

300 HITs will be posted on Amazon Mechanical Turk, and we therefore expect 300 responses. Any small deviation from this target will be because of the platform.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

We will observe people's choices throughout the game, and explore if the manipulation of the slope also exerts an impact on people's preference for exploration/exploitation. We will also measure people's beliefs about the predictive strength of the four attributes, and see if there is convergence/agreement on certain attributes being more diagnostic than others. Those analysis will not be reported as pre-registered.